

ENCHANT AGDM, AB, Canada

WMO: 715180

Lat: 50.1720N

Lon: 112.4366W

Elev: 2663

StdP: 13.34

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.7	-14.0	-25.3	1.5	-19.6	-19.5	2.1	-14.0	33.9	43.4	29.4	40.0	5.3	230	0.680

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.2	88.1	63.1	84.8	61.9	81.6	61.0	66.9	81.0	64.9	79.2	63.1	77.4	9.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.9	91.4	74.1	59.7	84.3	71.1	57.6	78.1	68.6	33.0	80.6	31.3	78.8	29.9	77.6	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.0	26.0	22.4	DB	-29.8	94.3	5.0	3.4	-33.4	96.7	-36.3	98.8	-39.1	100.7	-42.7	103.2	(4)
(5)			WB	-30.1	71.7	4.9	3.0	-33.6	73.8	-36.5	75.6	-39.2	77.2	-42.8	79.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.8	20.1	19.1	30.7	41.9	53.0	60.0	65.6	63.4	55.1	42.6	29.5	18.6
	DBStd	20.28	15.97	15.21	13.75	9.10	7.83	5.56	4.99	5.67	8.10	9.24	13.73	15.27
	HDD50	4612	928	867	599	263	59	2	0	1	41	260	618	974
	HDD65	8637	1392	1286	1063	692	379	168	53	97	309	695	1064	1439
	CDD50	1606	0	0	2	21	151	303	485	417	192	31	4	0
	CDD65	155	0	0	0	0	6	18	73	48	10	0	0	0
	CDH74	3237	0	0	0	21	226	380	1210	1008	380	12	0	0
	CDH80	1011	0	0	0	3	50	93	399	349	116	1	0	0
	WSAvg	9.5	10.5	8.9	10.4	10.8	10.2	9.3	7.6	7.4	8.5	9.7	10.5	9.8
	PrecAvg	13.3	0.5	0.4	0.6	1.1	1.9	2.8	1.5	1.5	1.1	0.7	0.6	0.5
(15) Precipitation	PrecMax	20.8	1.3	0.8	2.4	2.4	3.9	8.0	3.6	3.3	4.6	2.2	1.6	1.1
	PrecMin	6.5	0.1	0.0	0.0	0.4	0.3	0.8	0.0	0.1	0.0	0.2	0.0	0.0
	PrecStd	2.7	0.2	0.2	0.5	0.5	0.9	1.7	0.9	0.8	1.0	0.5	0.4	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.5	55.5	64.2	76.8	85.2	88.5	92.1	91.8	88.3	75.3	64.9	52.9
		MCWB	43.6	42.7	46.7	52.4	57.2	64.2	66.0	62.7	59.6	54.3	48.5	42.0
	2%	DB	47.9	48.6	59.6	69.7	80.4	82.2	88.2	87.7	83.5	69.0	57.2	47.6
		MCWB	39.6	39.3	45.2	49.2	56.4	61.0	65.7	62.1	58.6	51.7	45.3	39.1
	5%	DB	44.2	43.2	54.8	64.4	75.4	78.3	84.5	84.0	78.4	64.1	52.4	43.4
		MCWB	37.4	36.1	42.5	46.9	54.3	59.7	64.3	61.4	56.5	49.2	42.5	36.5
	10%	DB	40.5	39.2	49.9	59.1	70.6	74.5	81.0	79.9	72.8	59.1	47.9	39.8
		MCWB	35.3	33.8	39.9	44.4	52.1	57.5	63.0	60.8	54.8	46.4	39.3	34.4
	0.4%	WB	44.1	43.2	48.6	53.1	59.2	66.8	71.4	67.4	62.3	55.7	50.0	43.9
		MCDB	51.9	51.4	61.3	73.1	80.3	81.1	84.6	83.6	82.1	70.9	63.0	51.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.3	39.3	46.0	50.1	56.9	63.8	68.4	65.0	59.7	52.5	45.7	39.6
		MCDB	47.3	48.3	58.0	67.5	76.8	77.8	82.2	80.6	79.2	66.7	55.7	46.3
	5%	WB	37.9	36.5	43.2	47.5	55.1	61.2	66.2	63.3	57.6	49.7	42.8	36.9
		MCDB	43.5	42.6	53.6	62.1	72.4	74.6	79.8	79.2	76.3	62.1	51.7	43.0
	10%	WB	35.5	34.0	40.1	45.0	53.2	59.1	64.4	61.7	55.4	47.4	40.0	34.4
		MCDB	40.2	39.0	49.2	57.4	68.2	71.5	78.3	77.0	70.5	58.2	47.2	39.7
(35) Mean Daily Temperature Range	5% DB	MDBR	19.9	21.1	21.7	24.2	25.0	23.2	27.2	28.0	27.2	24.7	21.1	19.4
		MCDBR	20.9	25.6	29.1	35.5	34.5	31.0	33.6	36.9	38.4	32.6	26.3	21.1
		MCWBR	14.3	16.9	16.7	17.7	14.4	13.0	14.7	15.0	16.3	17.0	16.0	14.0
	5% WB	MCDBR	20.1	24.6	28.0	32.2	30.5	27.5	30.2	31.9	35.7	31.2	25.5	21.4
		MCWBR	14.4	17.2	16.8	16.7	13.1	12.9	14.7	14.3	15.6	16.9	16.2	15.0
(40) Clear-Sky Solar Irradiance	taub		0.260	0.269	0.292	0.331	0.347	0.343	0.344	0.362	0.320	0.288	0.262	0.245
	taud		2.449	2.481	2.465	2.360	2.364	2.425	2.437	2.365	2.458	2.524	2.494	2.446
	Ebn at Noon		256	283	293	290	288	290	287	276	279	271	254	246
	Edh at Noon		18	23	28	35	37	35	34	35	28	22	17	16
	RadAvg		389	703	1115	1536	1831	2016	2201	1757	1246	771	425	297
(45) All-Sky Solar Radiation	RadStd		31	62	86	105	138	134	110	111	110	83	37	27

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (77 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page